

Chapter 18

Adding and Organizing Media with the Media Pool

Before you can edit or grade media, you need to add it to the Media Pool, which is the central repository of clips in DaVinci Resolve. The Media Pool is a feature-rich environment, giving you many different methods of importing clips into your project and organizing them.

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Copying Media Using the Clone Tool

One of the few things you may want to do before you add media to your project is to clone all camera original media onto a safe set of backup volumes, for redundancy in case any one volume fails. Additionally, you should consider cloning all media to an off-site backup as well.

Whether you're on-set working as a DIT, or doing data ingest at a post facility, the Clone Tool in the Media page lets you safely and accurately copy media from SD cards, SSDs, or disk drives, to multiple destinations, with a checksum report (based on a choice of six checksum options) written to the root of each destination volume that verifies the absolute accuracy of the duplicate media saved to each destination.

To duplicate media using the Clone Tool:

- 1 Open the Clone Tool by clicking the Clone button at the far left of the Media Pool toolbar, which reveals the Clone Tool palette.
- 2 Click the Add Job button at the bottom left to create a new job. A job item appears within the Clone Tool palette, with overlays to guide you through its use.
- 3 Drag a volume or folder from the Media Storage panel to the "Drop source here" drop zone. Alternately, you can right-click any volume or folder in the Media Storage panel and choose Set As Clone Source.
- 4 Next, drag one or more volumes or folders from the Media Storage panel to the "Drop destination here" drop zone. Alternately, you can right-click any volume or folder in the Media Storage panel and choose Set As Clone Destination. You can have more than one destination.
- 5 If you want to preserve the top level folder name from the source volume or folder, click the Clone Tool panel's option menu, and choose "Preserve Folder Name." The overall folder structure of the cloned media is always preserved.
- 6 If you want to change the checksum method used by DaVinci Resolve to verify that each clip has copied properly, you can choose an option from the Checksum submenu of the Clone Tool's option menu. Each option is a tradeoff between the speed of your file copy operation and the security of the verification process. Greater security generally means a slower copy operation.

The options are:

None: Disables data verification, sacrificing safety for speed.

File Size: Fast, but minimal data verification. Data verification is done simply by comparing the file size of a duplicate file with that of the original. "Collision resistance" refers to whether two files (or a file and an incorrectly duplicated file) may coincidentally have the same comparison value (be it file size, an error-detecting code, or a hash). File Size is very fast, but it's minimally collision resistant.

CRC 32: Faster than MD5, but less secure. An error-detecting code rather than the hash used by the next three options. A "check value" is generated based on the remainder of a polynomial division of the file's contents. By comparing the check value derived from an original file with that derived from a copy, data integrity can be verified. This is a much faster data verification scheme than MD5 (the default), but it is significantly less collision resistant.

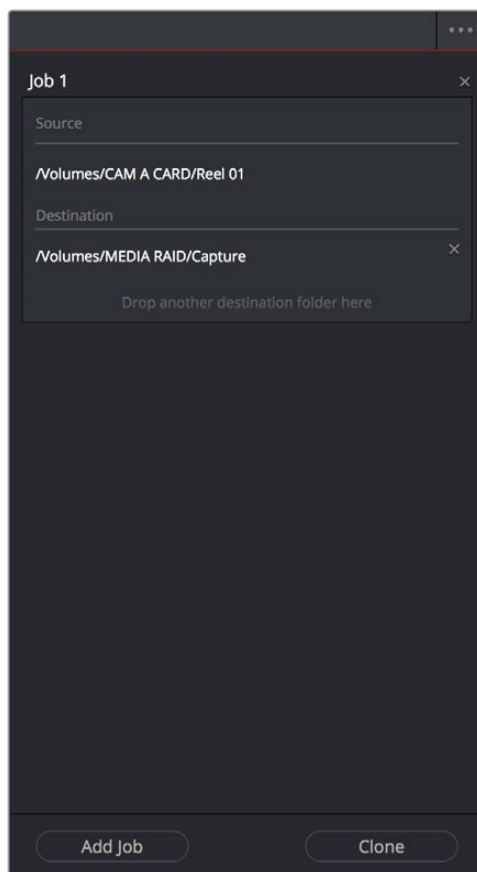
MD5: This is the default setting. A reasonable tradeoff between speed and security. A hash function generates a 128-bit value that's unique to a particular file; Data integrity is checked by comparing the hash value generated by the original file to that generated by the copied file. MD5 is not as collision resistant as the SHA options, but it's a faster operation, and the probability of such collisions in conventional film and video workflows is probably small.

SHA 256, SHA 512: Slower, but more secure. SHA is a more collision resistant hash function than MD5; options are provided for 256- and 512-bit value generation, with 512 being even more collision resistant than 256. However, these options are progressively slower than MD5, and will result in significantly slower copy times. Similarly to MD5, data integrity is checked by comparing the hash value generated by the original file to that generated by the copied file.

- 7 When you're ready, click the Clone button to initiate the cloning process.

To duplicate media quickly using the Clone Tool:

- 1 Right-click any volume or folder in the Media Storage panel, and choose Set as Clone Source. A job item appears within the Clone Tool palette, populated by the volume or folder you selected.
- 2 Next, right-click any volume or folder in the Media Storage panel, and choose Set As Clone Destination. You can do this more than once because you can have more than one destination.
- 3 If you want to preserve the top level folder name from the source volume or folder, click the Clone Tool panel's option menu, and choose "Preserve Folder Name." The overall folder structure of the cloned media is always preserved.
- 4 When you're ready, click the Clone button to initiate the cloning process.



The Clone tool with a job set up

Adding Media to the Media Pool

At minimum, you'll be using the Media page to add clips to a project to begin editing, in preparation to create dailies, or as a prelude to conforming a project using an EDL. All clips you want to work with must first be added to the Media Pool to be available for grading and processing in DaVinci Resolve, regardless of whether or not there's edited project data to go along with it.

If you import XML or AAF projects, you can choose to automatically import all accompanying media as part of the import process you initiate in the Edit page. However, if you find yourself needing to replace updated effects or stock footage in the Timeline, or you're called upon to add additional media such as animated titles or superimposed clips for compositing, then you'll still need to use the Media page to do so.

Whatever kind of project you're working on, you can add clips to the Media Pool from as many different volumes as you need. All imported clips are linked to the original media on whichever disks you found them; files are not moved, copied, or otherwise transcoded when you add them to the Media Pool. Consequently, it's a good idea to make sure that all media you want to import into your project has already been copied to a suitably fast volume before importing it.

Basic Methods for Adding Media in the Media Page

There are several ways of adding clips to the Media Pool.

To add individual clips from the Media Storage panel to the Media Pool:

- 1 Use the Media Storage panel to find a media file to import.
- 2 If you have multiple bins available in the Bin list, choose the bin you want to add the incoming media to.
- 3 Do one of the following:

Shift-click or Command-click multiple files, then right-click one of the selected files and choose "Add into Media Pool."

Drag a clip from the Media Storage panel browser to the Media Pool or to a specific bin in the Bin list.
- 4 If a dialog appears asking if you want to change your project to match the criteria, click "Change" to alter the project's settings, or click "Don't Change" to continue importing the media while leaving the project at its previous frame rate. Once clips have been imported into the Media Pool, the frame rate cannot be changed again, so choose carefully.

You also have the option of dragging media directly from the file system of supported platforms into the Media Pool.

To drag one or more clips from the File System to the Media Pool (supported platforms only):

- 1 Select one or more clips in your File System.
- 2 Drag those clips into the Media Pool of DaVinci Resolve or to a specific bin in the Bin list.

Those clips are added to the Media Pool of your project.

If you need to add the contents of all directories and subdirectories to the Media Pool as a flat group of media, that's easily accomplished. A good example of this is when you're importing camera original media from a cloned file structure, in which clips are organized into subdirectories that are many levels deep. DaVinci Resolve can easily import all of these clips and put them all into the same bin.

To add the entire contents of one or more directories of clips to the Media Pool:

- 1 Use the Media Storage panel to find and select one or more directories containing media files you need to import.
- 2 If you have multiple bins available in the Bin list, select the specific bin you want to add the incoming media to.
- 3 Do one of the following:

Right-click the selected directory or directories in the Media Storage panel, and choose "Add Folder into Media Pool" to add only clips from the selected directory. Subdirectories are ignored.

Right-click the directory in the Media Storage panel, and choose "Add Folder and SubFolders into Media Pool" to add clips from the selected directory and those from all subdirectories within.

Drag one or more selected directories you want from the Media Storage panel's browser area to the browser area of the Media Pool to add its contents, and the contents of all subdirectories within, to the currently selected bin in the Bin list.

You also have the option of using the directories and subdirectories that organize media in your file system as bins in the Media Pool, so that you can preserve the original organization of your media.

To add all clips and folders in a directory organized into matching folders in the Media Pool:

- 1 Use the Media Storage panel to find the directory containing the files you need to import.
- 2 Do one of the following:
- 3 Right-click the directory and choose "Add Folder and SubFolders into Media Pool (Create Bins)"
- 4 Drag the folder you want to import from the Media Storage panel to the Bin list of the Media Pool to add that folder, and all subfolders within, as a new bin in the Bin list.

A folder appears in the Media Pool with the same name as the folder you dragged in. All clips and all subdirectories appear within, nested hierarchically in the Media Pool as they were in the file system.

Import Hierarchically Organized Nests of Empty Directories

You can also import a nested series of directories and subdirectories that constitutes a default bin structure you'd like to bring into projects, even if those directories are empty, by dragging them from your file system into the Media Pool Bin list of a project. The result is a hierarchically nested series of bins that mimics the structure of the directories you imported. This is useful if you want to use such a series of directories as a preset bin structure for new projects.

Adding Subclips From the Media Storage Panel

If you're browsing long source clips in the Media Storage panel, but you only want to import a small segment of a much longer clip into the Media Pool, you can create subclips directly from the Media Storage panel.

To add a subclip from a clip in the Media Storage panel to the Media Pool:

- 1 Single-click any clip in the Media Storage panel to open it into the Viewer in order to create a subclip without needing to first import that clip into the Media Pool.
- 2 Set In and Out points in the Source Viewer to define the section you want to turn into a subclip.
- 3 Do one of the following:
 - Right-click the jog bar and choose Make Subclip from the contextual menu
 - Drag the clip from the Viewer to the Media Pool to add it as a subclip

Adding Individual Frames From Image Sequences

If you're working with image sequences, or with sequentially numbered image files from any source, DaVinci Resolve automatically presents them as clips in the Media Storage panel. This is good if that's what they are, but there are instances where sets of photos, of which each frame is in actuality a separate media file, are also sequentially numbered. For this reason, you can import individual frames, rather than entire image sequences.

To choose between adding individual frames from a number sequence of images, or adding them as image sequence clips in the Media Storage panel:

- 1 Click the Media Storage panel option menu, and choose Frame Display Mode.
- 2 Close one of the drop-down options:

Auto: DaVinci Resolve will automatically select Individual Frames or Image Sequence based on file type. For example, DPX and EXR files will be imported as image sequence clips, while JPG files will be imported as individual frames.

Individual: Each image sequence is now separated into its individual frames, allowing you to select only the frames you need.

Sequence: Will group sequentially numbered files together as an image sequence clip, regardless of file type.

- 3 Use any of the previous described methods to add the frames you want to the Media Pool as individual clips or image sequences.

Adding Media Based on EDLs

Another strategy for adding media to the Media Pool is to use an EDL to add only the clips it refers to from a directory. This lets you add only the clips that are necessary for conforming a particular imported project before conforming an EDL, and eliminates the need to add too much media to the Media Pool, which might slow you down in the case of projects referencing terabytes of media. Furthermore, you can choose multiple EDLs to base the import on, and many directories to examine.

The EDLs will reference clips via their timecode and sometimes reel name and path. It is these settings and the conform frame rate that you made previously in the Configuration screen that are now used to place images correctly into the Media Pool.

To add only media used in an EDL to the Media Pool:

- 1 If necessary, open the General Options panel of the Project Settings, turn on the "Assist using reel names from the" checkbox, and choose a method with which to extract reel name information from the media files you're about to import.

For more information, see *Chapter 19, "Using Clip Metadata."*

- 2 Right-click a directory in the Media Storage panel, and choose one of the following commands:
 - Add Folder Based on EDLs into Media Pool
 - Add Folder and SubFolders Based on EDLs into Media Pool
- 3 Using the file dialog that appears, select one or more EDLs to use.

DaVinci Resolve searches the directory hierarchy, either one level deep or all levels deep, for every media file matching the source timecode and the reel ID of an event in one of the selected EDLs.

Splitting Clips Based on EDLs

You can also use EDLs to split a media file into multiple clips in the Media Pool, either as an alternate means of “preconforming” a flattened master media file, or to import multiple sections of a longer media file that happen to be referenced by an EDL.

To split and add clips based on an EDL:

- 1 Right-click a directory in the Media Storage panel, and choose “Split and add into Media Pool.”
- 2 Using the file dialog that appears, select an EDL to use, and click Open.
- 3 Choose a frame rate to use to conform the clips to in the “File Conform Frame Rate” dialog, and click OK.
- 4 Choose a handle size, in frames, and whether or not you want to split unreferenced clips from the “Enter handle size for splitting” dialog, and click Split & Add. The media file is split into the component clips specified in the EDL, and added to the Media Pool.

TIP: Turning on the Split Unreferenced Clips checkbox automatically splits out sections of the file that were not referred to by the EDL you selected, and adds them to the Media Pool separately, giving you access to every piece of media that’s available.

Import Clips With Metadata Via Final Cut Pro 7 XML

In order to support workflows with media asset management (MAM) systems, DaVinci Resolve supports two additional Media Pool import workflows that use Final Cut Pro 7 XML to import clips with metadata.

To import clips with metadata using Final Cut Pro 7 XML files, do one of the following:

Right-click anywhere in the background of the Media Pool, choose Import Media from XML, and then choose the XML file you want to use to guide import from the import dialog.

Drag and drop any Final Cut Pro 7 XML file into the Media Pool from the macOS Finder.

Every single clip referenced by that XML file that can be found via its file path will be imported into the Media Pool, along with any metadata entered for those clips. If the file path is invalid, you’ll be asked to navigate to the directory with the corresponding media. Additionally, the following metadata is imported:

- Clips
- Browser metadata
- Subclips
- Clip Markers, with colors and duration
- Bin Hierarchy
- Comments

Adding Media With Offset Timecode

Sometimes source media was created with incorrectly offset timecode, due to mistakes made earlier in the postproduction process. If this offset is consistent, you can use the “Add Folder with Source Offset” command to add media to the Media Pool as clips with a timecode offset.

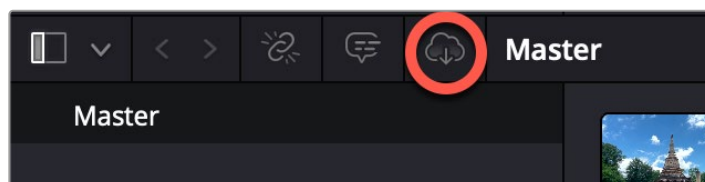
To add a folder of clips to the Media Pool with offset timecode:

- 1 Right-click a directory in the Media Storage panel, and choose one of the following commands:
 - Add Folder with Source Offset
 - Add Folder and SubFolders with Source Offset
- 2 Choose a number of frames with which to offset the timecode from the “Change Frame Offset” dialog, and click Apply.

The media is imported as clips with offset timecode in the Media Pool. However, the original source timecode of the clips on disk has not been altered. All media rendered out of the Deliver page will reflect the offset timecode.

Import Blackmagic Cloud Shared Folders to Media Pool

You can now import and sync Blackmagic Cloud Folders into the Media Pool. This allows you to connect to one or more cloud media folders and selectively download media from them to your local machine. Blackmagic Cloud Folders are online common storage folders that are not tied to a specific project. This lets you create a pool of commonly used media assets, such as title sequences, credits, bumpers, etc., and share them online without having to import them separately into each individual project.



The Import Cloud Folder icon at the top of the Media Pool

To link a Blackmagic Cloud Folder to your project:

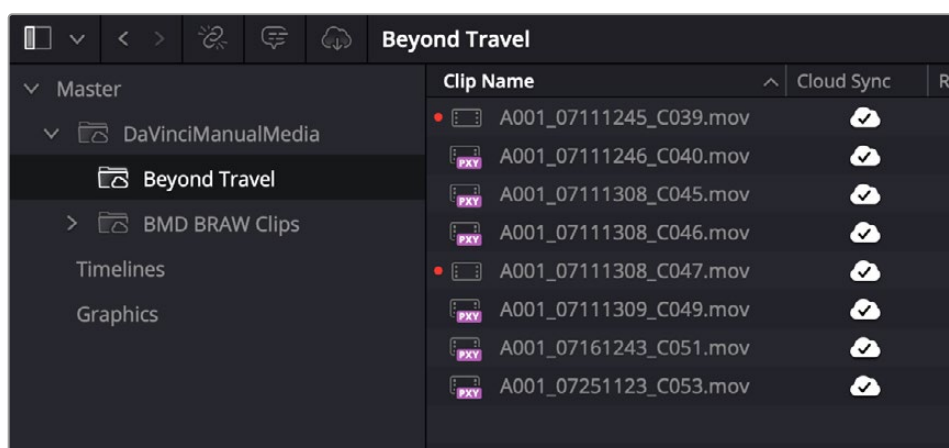
- 1 Sign into your Blackmagic Cloud account in DaVinci Resolve.
- 2 Click on the Import Blackmagic Cloud Folder icon at the top of the Media Pool in any page.
- 3 In the Blackmagic Cloud Folder dialog, click select to choose a Blackmagic Cloud Folder. The choices here will show any Blackmagic Cloud Folder that you have created or someone else has shared with you. Click the Add button.
- 4 In the Download Folder Location, select a file location on your local system that you want the media from the Blackmagic Cloud Folder to be stored in.
- 5 Choose whether you want to sync proxies only, or proxies and used originals.
- 6 Press Download to add the Cloud Folder to your Media Pool.



The Blackmagic Cloud Folder link dialog

When you open the Blackmagic Cloud Folder in the Media Pool, it will be empty and populate one clip at a time as their clip names (not their media) are downloaded from the cloud. Also, the Blackmagic Cloud Folder is not actually linked to your project yet; it is only linked when the first media clip is “used.”

A “used” clip is new terminology in DaVinci Resolve and simply means a clip that has been altered in any way inside the project. A clip can be used by putting it on a timeline, certainly, but also includes other actions like applying a flag, altering its metadata, transcribing it, etc. Once a clip is used, its media will start to download to your local machine. Clips that are used are denoted by a red dot next to them in the Media Pool. Any unused clips will remain as virtual clips in the Cloud Folder until used. Once downloaded, clips in Cloud Folders can be used just like any other clip in the Media Pool.



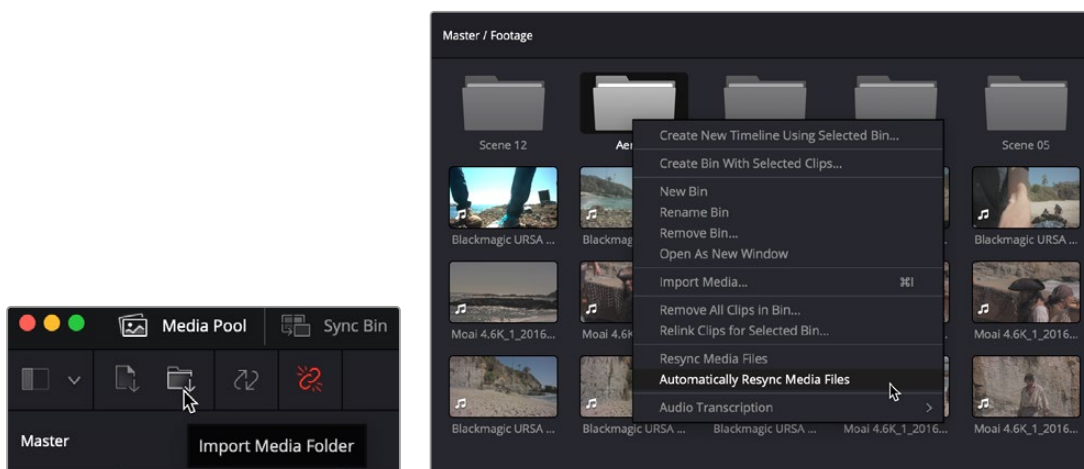
The Cloud Folder linked in List view. Clips with red dots are “used,” and the original media has been downloaded. Unused clips have just their proxies available to view. The Cloud Sync column shows that the media is registered and synced with the Blackmagic Cloud.

Sync Media Pool Bins with File System Folders

You can sync a Media Pool bin with a folder on your computer's file system (MacOS, Windows, Linux, etc.). This allows you to add additional media into a folder on your computer and have it automatically import to its respective bin in DaVinci Resolve.

To set up a folder with Automatic Syncing:

- 1 On the Cut page, use the Import Folder icon on the top menu bar to set up matching folder and sub folder names.
- 2 Right-click on the imported folder and choose "Automatically Resync Media Files."

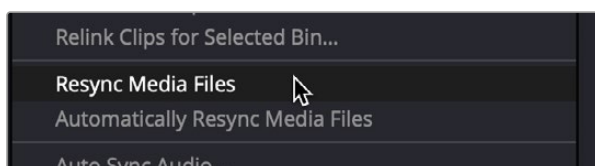


Use the Import Media Folder icon in the Cut page

Right-click on the bin and select Automatically Resync Media Files

Newly added clips in this folder will be added automatically to the media bin. This can be useful when working with cloud-synced media files (as when using DaVinci Resolve's Replay capabilities), or any files that may be added to the folder later.

To manually resync the contents of a Media Pool bin once, right-click a bin and choose "Resync Media Files."



Right-click on the bin and select Resync Media Files to perform the sync operation manually.

For both Resync actions, ensure that:

- At least one clip is present in the media bin.
- All existing clips in the media bin share the same file path.
- The media bin name matches the parent folder of the clip paths. You can ensure this is set up correctly by using "Import Media Folder" in the Cut page, or "Add Folder and SubFolders into Media Pool (Create Bins)" in the Media page.

Sync actions will add new clips from sub-folders if the corresponding sub-bins exist. However, newly created sub-folders without corresponding media bins are not tracked, and new media bins without the corresponding sub-folders are skipped.

For example, say you were editing a sporting event, and you had a graphics operator creating animated lower-thirds for each of the players. By creating a common network folder called “CG” and automatically syncing it in DaVinci Resolve, each time a new lower-third was finished and saved, it would automatically appear in your bin in the Media Pool ready for use.

Here’s how you would set this up:

- Create a folder on your network storage called “CG.” Both the editor and the graphics operator need to have access to this folder.
- Note that because this folder is currently empty, it can not be used in a Resync action yet. Only after there is at least one media file in the folder can you perform a Resync.
- The graphics operator finishes his first animated lower third and saves it to the “CG” folder as a .mov file (though any DaVinci Resolve readable media format would work).
- The editor then imports the “CG” folder into the Media Pool in DaVinci Resolve and selects Automatically Resync Media Files from the context menu.
- As the graphics operator finishes and saves additional lower thirds into that folder, they automatically appear in the Editor’s “CG” bin in the Media Pool, without having to specifically import them.
- For organizational purposes, the graphics operator then creates a new folder in “CG” for the other team, called “Team B.” Unfortunately, because this folder structure was not in the original imported bin, it does not show up in the editor’s media pool. However, if they had created this Team B folder before the editor imported it, and it had media in it, it would be there.
- Now in this case, the editor simply imports the same bin again, selects Automatically Resync Media Files, and the new folder and all its media appear as expected. Any further files in dropped in “Team B” will show up in the editor’s media pool.

IMPORTANT: Media entries for clips that are removed or not found on the file system are set as offline media. This allows for retention of metadata and is useful when the clip path is temporarily offline, on cloud storage where file status may be in flux, or moved away by apps or workflows seeking to write a new version in its place.

Adding Media to the Cut, Edit, Fusion, and Fairlight Pages

While adding clips to the Media Pool in the Media page provides the most organizational flexibility and features, if you find yourself in the Cut, Edit, Fusion, or Fairlight page and you need to quickly import a few clips for immediate use, you can do so in a couple of different ways.

To add media by dragging one or more clips from the Finder to the Media Pool (macOS only):

- 1 Select one or more clips in the Finder.
- 2 Drag those clips into the Media Pool of DaVinci Resolve, or to a bin in the Bin list.

Those clips are added to the Media Pool of your project.

To use the Import Media command in the Media Pool:

- 1 Right-click anywhere in the Media Pool, and choose Import Media.
- 2 Use the Import dialog to select one or more clips to import, and click Open.

Those clips are added to the Media Pool of your project.

Removing Media From the Media Pool

If you've added clips to the Media Pool that you need to eliminate, this is easy to do, either singly, or in the aggregate.

To remove clips from the Media Pool, do one of the following:

- Select one or more clips in the Media Pool, then press the Delete or Backspace key.
- Select one or more clips in the Media Pool, right-click one of the selected clips, and then choose Remove Selected Clips.
- Right-click anywhere in the Media Pool, and choose Remove All Clips in Bin.

NOTE: If you've turned on "Automatically match master timeline with media pool" in the General Options panel of the Project Settings, you cannot remove all clips from the Media Pool if there are other timelines using that media.

To remove clips from the Master Timeline (if it's exposed):

- Open the Edit page, then select one or more clips in the Media Pool, right-click one of the selected clips, and choose "Remove Selected Clips from Master Timeline."

For more information about using the Master Timeline, see *Chapter 33, "Using the Edit Page."*

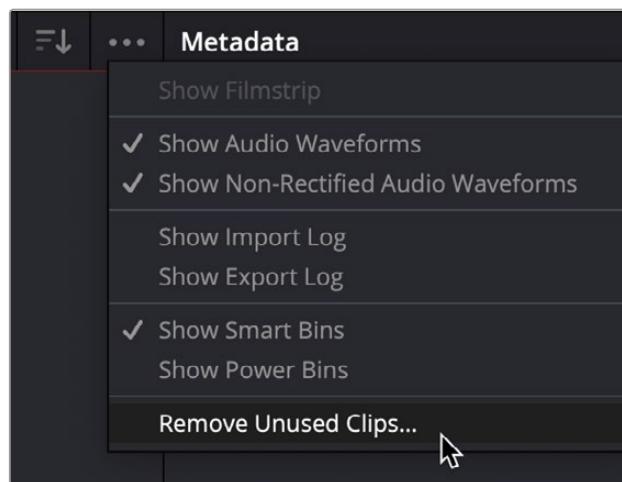
Removing Unused Media from a Project

You can have DaVinci Resolve automatically remove any unused media clips from the Media Pool in preparation for archiving or handing off a clean project to another workstation. This action loads all timelines, compound clips, and Fusion compositions then analyzes them for media usage. Any clips not found during this analysis are removed from the Media Pool. This option does not delete any clips from your disk, it only removes them from the Media Pool and the Project. This operation is un-doable.

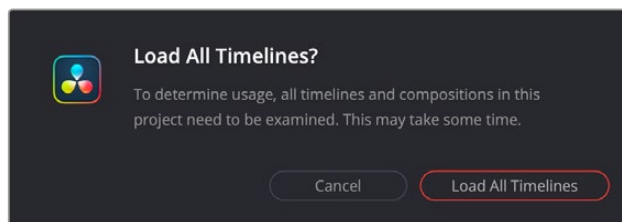
Depending on the size of the Project, this operation can take several minutes.

To remove unused media from a Project:

- 1 Click on the Media Pool Option (3-dot) menu.
- 2 Select Remove Unused Clips from the drop-down menu.
- 3 Click on the Load All Timelines button to start the operation, or Cancel to abort it.



Select Remove Unused Clips from the Media Pool Option menu



Click the Load All Timelines button to start the analysis

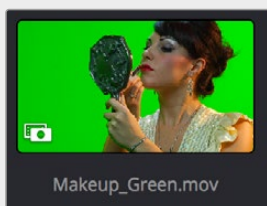
Adding and Removing External Mattes

If you've been provided with matte files to accompany one or more media files used by a program you're grading, you can attach them directly to specific clips in the Media Pool, in order to use them as key sources for a Clip Grade in the Node Editor of the Color page. You can even use matte files that pack multiple mattes within a single piece of media. This can be done by either writing different mattes to each of the red, green, and blue channels of a clip, or by embedding multiple matte passes within a single OpenEXR file.



Matching RGB and Matte images

When the Media Pool is in Icon view, clips with clip mattes appear with a badge.



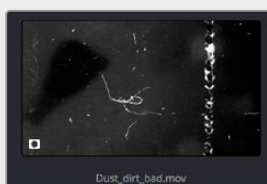
A clip matte, seen in Icon view

Clip mattes appear listed underneath a clip in the Media Pool when it's in List view.

Clip Name	Start TC	End TC
 Makeup_BG.mov	07:10:54:04	07:11:10:10
▼  Makeup_Green.mov	07:10:54:04	07:11:10:10
 Makeup_Matte.mov	07:10:54:04	07:11:10:10

A clip matte, seen in List view

Alternately, you can add a timeline matte to the Media Pool, which isn't attached to any clip, that can be used as a key source in the Color page within any clip's Clip grade, or within a Timeline Grade. Timeline mattes appear as stand-alone clips in the Media Pool.



A timeline matte,
seen in Thumbnail view

What Are Mattes For?

Matte files are useful for two things. Traditionally, mattes are grayscale media files that identify regions of varying opacity, with white representing solid areas, and black representing transparency. For example, exported clips from a compositing application sometimes are accompanied by one or more matte files that correspond to keys or rotoscoped mattes from the composite. By importing these matte files using the “Add as Matte” command, you can attach them to the clips they belong to in the Media Pool, so that they’re only available to the clips they’re synced to.

However, mattes can also be used as creative tools to apply grain and texture for effect. What a matte does depends on how you connect it in the Node Editor of the Color page. These are media files that you may want to use as mattes for potentially any clip, so they can also be added to the Media Pool as a so-called timeline matte, that can be applied to any clip you want.

TIP: If necessary, you can also apply LUTs to both clip mattes and timeline mattes in the Media Pool, simply by right-clicking a matte, and choosing a LUT from the 1D LUT or 3D LUT submenus. This can be helpful for adjusting incorrectly formatted mattes.

Adding Mattes

To use mattes, you need to add them in very specific ways.

To assign a matte to a clip in the Media Pool:

- 1 Select a clip in the Media Pool to which you want to attach an external matte.
- 2 Select the matching external matte file in the Media Storage panel, right-click it, and choose Add to Media Pool as a Matte.

The matte is attached to the clip as a clip matte. A badge indicates that clip has a matte when the Media Pool is in Icon view, and the matte itself can be seen, if you put the Media Pool into List view, appearing as a nested item underneath the clip it’s attached to.

Removing mattes from clips in the Media Pool:

- 1 Put the Media Pool into List view.
- 2 Right-click the external matte file you need to remove, and choose Remove Selected Clips.

Removing an external matte clip also removes that matte’s key from any clip grades that use it, such that any clips using it as a key input change from a secondary operation to a primary operation, where the color adjustment affects the entire image.

To add a timeline matte to the Media Pool:

- 1 Make sure no clip is selected in the Media Pool.
- 2 Select an external matte file in the Media Storage panel, right-click it, and choose Add to Media Pool as a Matte.

The external matte appears in the Media Pool as a timeline matte.

You can also assign mattes to clips directly in the Color page, which can sometimes be faster.

To assign a matte to a clip in the Color page:

- Drag any clip from the Media Pool to the Node Editor.

That clip appears as an External matte for the current clip's grade in the Node Editor, and it's also automatically assigned to the current clip in the Media Pool.

For more information on using external matte clips as keys when grading, see *Chapter 146, "Combining Keys and Using Mattes."*

Using Embedded Mattes in OpenEXR Files

If you're importing OpenEXR files with embedded matte passes, there's nothing special you need to do, as the mattes are within the clip you've just imported into the Media Pool.

For more information on how to use mattes within OpenEXR files, see *Chapter 146, "Combining Keys and Using Mattes."*

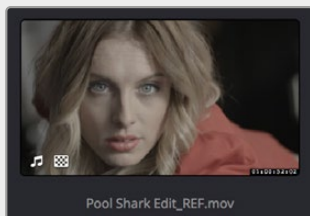
Adding Offline Reference Movies

When moving a project from another application to DaVinci Resolve, it's useful to export the entire program as a single media file for use as an Offline Reference Movie. Then, you can import this file in a special way to use for dual Viewer comparison in the Edit page, or as a split-screen comparison for fade wipe in the Color page. As of DaVinci Resolve 16 it's no longer necessary to import reference movies in this way to make an offline comparison, but it can still be convenient when managing multiple timelines and versions that require great specificity.

To add a clip as an offline reference clip:

Right-click it in the Media Storage panel, and choose "Add As Offline Clip."

That clip appears with a small checkerboard badge in its icon in the Media Pool, or as the icon to the left of the Media Pool.



Checkerboard icon indicating an Offline comparison video

For more information on using an offline video to compare with an imported Timeline in the Edit page, see *Chapter 55, "Preparing Timelines for Import and Comparison."*

For more information on split-screen reference of Offline video in the Color page, see *Chapter 124, "Using the Color Page."*

Extracting Audio in Media Storage

If there's a video clip in the Media Storage panel that has audio you need, but you don't want the video component, you can use the Extract Audio command to create a self-contained audio clip that you can then import into the Media Pool by itself.

To extract the audio from a media file:

- 1 Right-click a clip in the Media Storage panel, and choose Extract Audio.
- 2 Click the Browse button in the Extract Audio dialog to find another disk location for the extracted clip.
- 3 Click Extract. The audio channels are extracted and written as a .WAV file to the selected destination.
- 4 After you've extracted the stand-alone .WAV file, you'll need to import it into the Media Pool if you want to use it in your project.

Manually Organizing the Media Pool

Whether you're doing onset work, creating digital dailies, organizing media to edit, or ingesting media to conform to an imported project, it's vitally important to stay organized. The Media Pool provides many different tools for doing so. This section examines how you can create bins to manually organize collections of clips.

To Select Clips in the Media Pool

There are a variety of ways you can make clip selections in the Media Pool in preparation for relinking, unlinking, moving, duplicating, deleting, or doing any other operation to them.

- Click any clip to select it.
- Drag a bounding box around several clips to select them all.
- Hold the Command or Shift keys down and drag a bounding box around another discontinuous group of clips to either add them to the current selection or remove them from the current selection.
- Click one clip, then Shift-click another to select both clips and make a continuous selection of all clips in-between. Shift-clicking another clip can expand or contract the selection.
- Command-click individual clips to select a discontinuous number of clips. Command-click a clip that's already selected to individually de-select it, while leaving the rest of the selection alone.
- With one clip selected, hold the Shift or Command keys down and use the Arrow keys to expand the selection to other clips.

Organizing Media into Bins

You can easily organize clips into different bins in the Media Pool. For some workflows, this is required, while with other workflows it's purely optional.

Methods of working with bins in the Media Pool:

- **To add a bin to the Media Pool:** Right-click in the Bin list and choose Add Bin. To add a bin inside another bin, right-click any bin and choose Add Bin.
- **To move selected clips into a new bin:** Select all the clips you want to put into a new bin, then right-click one of the selected clips, and choose Create Bin With Selected Clips.
- **To rename a bin:** Select the bin you want to rename, and then click its name a second time to make it editable. With the bin name highlighted, type a new name and press Return. Alternately, you can right-click a bin, choose Rename Bin, and then type a new name and press Return.
- **To add incoming clips to a specific bin in the Media Pool:** Click a bin to select it, then use any of the previously described methods to add media from the Media Storage panel directly to that bin.
- **To move media from one bin to another:** Drag one or more selected clips from their current location in the Media Pool into that bin. Multiple clips in the Media Pool can be selected by Shift-clicking or Command-clicking them, or by dragging a bounding box over a group of clips. You can also drag one bin into another one.
- **To delete a bin:** Select the bin you want to delete, and press the Backspace or Delete key. Or, right-click a bin and choose Delete Bin. Deleting a bin with nested bins inside of it results in that entire set of bins being deleted.
- **To sort bins:** Right-click on any bin, and choose an option from the Sort By submenu. You can choose from Name, Date Created, Date Modified, and User Sort.
- **To reorganize bins manually:** Right-click anywhere within the Bin list, and choose Sort By > User Sort. Then, drag bins up or down in the Bin list to put them into the order you want. An orange dividing line shows where dragged bins will be placed when you drop them and helps you see when a bin you're dragging will become nested within another bin or not. The User Sort order is saved even when you change to another sort order, and selecting User Sort again results in your custom sort order being recalled.

Import and Export DaVinci Resolve Project Bins (.drb)

You can import/export specific bins from one DaVinci Resolve project to another, allowing you to pass bins quickly between projects and workstations that have access to the same media. All Metadata, In/Out points, Timelines, etc. are transferred along with the clips in the bin, but none of the actual media files are included.

To export bins from the Media Pool:

- 1 Select one or more bins in the Media Pool.
- 2 Right-click the selection and choose “Export Bin,” or choose File > Export > Export Bin.
- 3 Choose where to save the DaVinci Resolve Bin file (.drb) in the file system dialog, and click Save.

To import bins into the Media Pool:

- 1 Right-click in the Media Pool and choose “Import Bin,” or choose File > Import > Import Bin.
- 2 Do one of the following:
 - Choose a DaVinci Resolve Bin file (.drb) from the file system dialog.
 - Double click the .drb file in your file system.

The bin or bins will appear in the Media Pool. Any bins imported this way will have the word “import” appended to their name, to avoid duplicate names. If you import a bin that contains clips that were already in the Media Pool, the potentially duplicate clips are excluded from the import and instead relinked to the media referenced by your project. This keeps your Media Pool tidy. However, if the bin or bins have been moved to another computer, you may have to relink offline media.

Import and Export Power Bins

You can import and export Power Bins (bins that persist from project to project) as .drb files, just like normal bins. Power Bins are hidden by default and can be turned on by clicking on the Media Pool Option (3-dot) menu and selecting Show Power Bins.

Import and Export DaVinci Resolve Timelines (.drt)

You can export and import individual timelines from one DaVinci Resolve project into another previously existing DaVinci Resolve project, allowing you to pass timelines quickly between projects and workstations, without creating additional imported project files. Just the timeline and its associated clip information is exported, none of the actual media files are included.

To export a timeline from the Media Pool:

- 1 Select a timeline from the Media Pool.
- 2 Choose File > Export > Export AAF, XML, DRT (Shift-Command-O).
- 3 Choose “DaVinci Resolve Timeline Files (*.drt)” from the format options popup in the file system dialog.
- 4 Choose where to save the DaVinci Resolve Timeline file (.drt) in the file system dialog, and click Save.

To export multiple timelines from the Media Pool:

- 1 Select multiple timelines from the Media Pool.
- 2 Right-click on a timeline and choose Timelines > Export > DaVinci Resolve Timeline Files.
- 3 Select the folder where you wish to save them from your file browser.

Unlike exporting individual timelines, when exporting multiple timelines you will not be able to rename the .drt files on export. They will retain the timeline name they have in the Media Pool.

To import a timeline into the Media Pool:

- 1 Choose a bin in the Media Pool in which you want the imported timeline to be saved.
- 2 Do one of the following:
 - Choose File > Import Timeline > Import AAF, XML, DRT (Shift-Command-I), then Select a DaVinci Resolve Timeline file (.drt) from the file system dialog, and click Open.
 - Double click the .drt file in your file system.

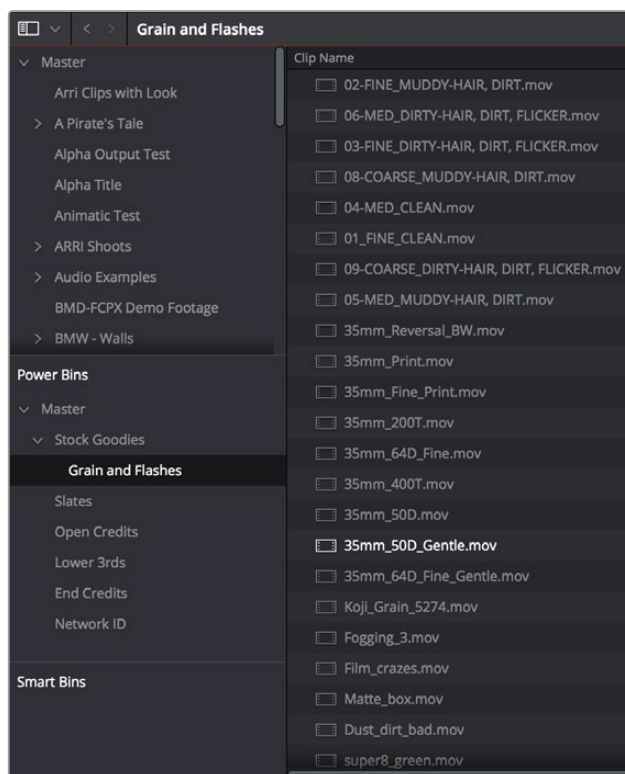
The timeline will appear in the Media Pool, along with all of the clips associated with it, including any media sync information. Any timelines imported this way will have the word “import” appended to their name, to avoid duplicate names. The imported timeline will be automatically conformed to corresponding media that’s already in the Media Pool. However, if the timeline has been moved to another computer, you may have to reimport or relink missing or offline media in to bring the imported timeline fully online.

NOTE: Only a single timeline can be imported and exported at a time using this method. To import or export multiple timelines, use the Import/Export Bin function described above.

Sharing Media Among Projects Using Power Bins

Power Bins provide a way of importing and organizing media that you want to be available to all projects in DaVinci Resolve. Power Bins reside in a separate area of the Media Pool, with resizable dividers separating them from both the ordinary bins and Smart Bins areas. Power Bins are hierarchical, just like regular bins, and you can nest as many as you like, one inside another.

Like regular bins, Power Bins must be manually created by right-clicking within the Power Bins area and choosing Add Bin. The difference is that whatever clips you import into Power Bins are shared among all projects in a single-user installation, or all projects belonging to a particular user in a multi-user installation. In this way, they’re similar to Power Grades in the gallery of the Color page. This makes Power Bins ideal for storing shared media that’s re-used often, such as stock video, sound effects, stills, and things like company slates and network graphics and animations that go into every show of a series.



The Power Bins area of the Bin list

Power Bins are created and used like any other bin, using the procedures described previously.

To show or hide the Power Bin area of the Bin list:

Choose Show Power Bins from the Media Pool option menu to toggle the visibility of all power bins on and off.

Automated Organization Using Smart Bins

A completely automated way of organizing media in the Media Pool is to use Smart Bins that are either automatically or manually created, in order to collect all clips and timelines in the Media Pool that have commonalities based on any of the intrinsic or user-editable metadata that's available in the Metadata Editor and Media Pool. If you're familiar with the Color page, Smart Bins work much the same way as Smart Filters, and they're created and edited using much the same procedures.

For more information about Smart Filters, see *Chapter 124, "Using the Color Page."*

Smart Bins are incredibly flexible. Using one or more metadata-based rules, they can be as simple or sophisticated as you require. They're even capable of using multiple groups of multiple rules for situations where you need to gather clips that match all of one set of criteria, but only one of a second set of criteria. In this way, you can use Smart Bins to solve a wide variety of organizational needs as you edit your program.

Smart Bins Are Only As Good As Your Metadata

It's important to point out, however, that as much intrinsic metadata is available to every clip in DaVinci Resolve automatically (clip properties such as frame rate, frame size, codec, file name, and so on), the more time you take entering extra metadata in the Metadata Editor to prepare your project for editing and grading, the more powerful Smart Bins can be in helping you to sift and sort through the contents of a program you're grading. Examples of metadata entry that will guarantee immediate benefits from Smart Bins include the entry of scene, shot, and take information, keywords identifying key descriptors (day and night, interior and exterior, framing, and so on), and using Face Detection to assign character names. These categories of metadata can be used for the automatic creation of Smart Bins, but they can also be used in combination when manually creating Smart Bins that are even more specific.

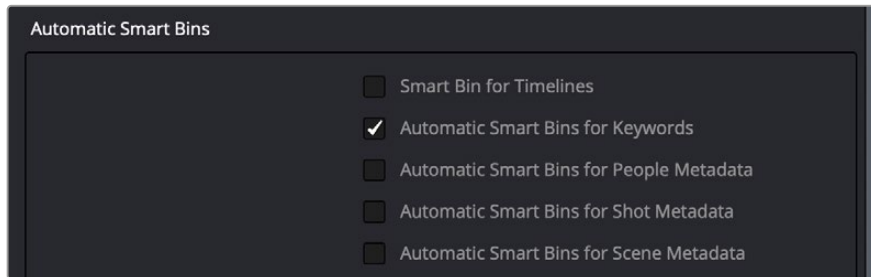
Imagine being able to gather all the clips in a particular scene, find all the interview clips for a particular subject, or find all the edited timelines corresponding to a particular name, all by simply selecting a Smart Bin that automatically examines the current contents of the Media Pool. If you or an assistant can take the time to enter metadata for the source material in a project that identifies these characteristics, you'll be able to work even more quickly to find the clips you need for any given situation.

Smart Bins Update Their Contents Dynamically

Smart Bins are always dynamically up to date and include whatever new media is added to the Media Pool. This makes it easy to stay organized, even when working on projects where new media is being added to the Media Pool every day, such as when editing during a shoot. By using metadata entered either in-camera, by the DIT or media wrangler managing ingest, or by an Assistant Editor who's working with you, Smart Bins will automatically include all clips in the Media Pool that have matching criteria, whether they were added a month ago or a minute ago.

Automatic Smart Bin Creation

The process of adding metadata to your clips can be used for the automatic creation of sets of “Smart Categories,” which are Smart Bins that are generated and organized by the presence of specific categories of metadata and appear in the Smart Bins section of the Media Pool sidebar. To enable or disable this behavior, open the Editing panel of the User Preferences, and use the checkboxes in the Automatic Smart Bins group to choose which metadata automatically creates Smart Bins.

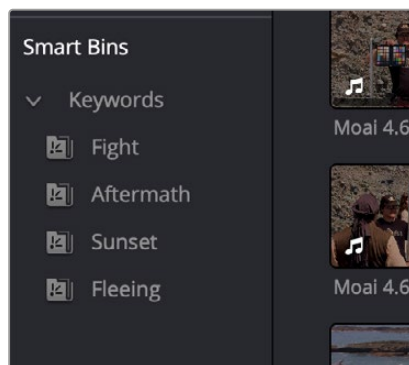


Preferences governing what metadata can automatically create Smart Bins

Metadata capable of creating Smart Bins include:

- Clip Keywords
- Marker Keywords
- People Keywords (added via People Detection)
- Scene metadata
- Shot metadata

These categories are hierarchically organized, with each category closed by default to save space. Click the disclosure triangle of any category to reveal all Keyword, People, Scene, or Shot Smart Bins that are available in the current project. Selecting the Smart Category's top bin lets you see every clip referenced by every Smart Bin inside of it, whereas selecting individual Smart Bins shows you only the clips referenced by that Smart Bin.



A Smart Category seen in the Smart Bins area of the Media Pool sidebar

Drag and Drop Clips to Assign Automatic Smart Bin Properties

Metadata entry does not need to be a one way process, starting with typing in the metadata editor. For certain Automatic Smart Bin categories (keywords, shot, and scene), you can drag multiple clips from the Media Pool on top of an existing smart bin to assign the properties of that bin to all the clips at once.

To automatically assign Smart Bin Properties to a range of clips:

- 1 Create a Smart Bin in one of the following categories: Keywords, Shot, or Scene.
- 2 Select all of the clips that you want to apply the Smart Bin property to.
- 3 Drag those clips from the Media Pool and drop them on top of the Smart Bin.

For example, you could make a smart bin with the Keyword: “Sunset.” That bin would automatically show up under the Keywords category in the Smart Bins. Then you could select all the sunset shots in your Media Pool, and drag them on top of that bin to apply the “sunset” keyword to all the clips at the same time.

NOTE: Dragging and dropping clips to assign Smart Bin properties only works with the Automatic Smart Bin Keywords, Shot, or Scene categories. Dragging clips to any other Smart Bin will have no effect.

Manual Smart Bin Creation

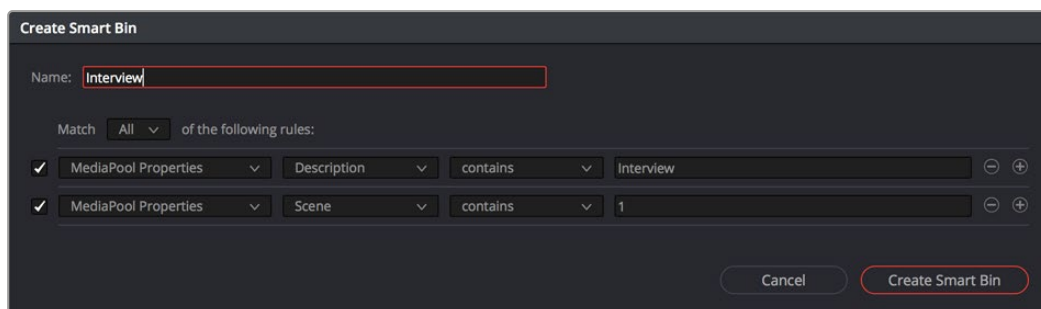
It's easy to manually create Smart Bins with customized rules to filter very specific collections of media and timelines that you want to use.

To show or hide the Smart Bin area of the Bin list:

- Choose Show Smart Bins from the Media Pool option menu to toggle the visibility of all Smart Bins on and off.

To create a Smart Bin:

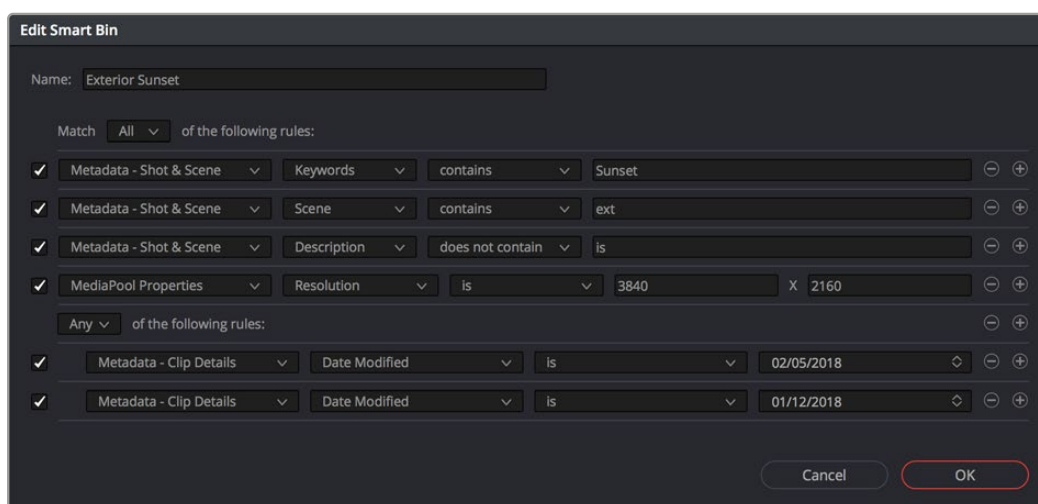
- 1 If necessary, open the Bin list, choose Show Smart Bins from the Media Pool option menu, then right-click anywhere in the background of the Smart Bin area of the Bin list, and choose Create Smart Bin.
- 2 In the Create Smart Bin dialog, enter a name for the filter, and use the following controls to create one or more filter criteria (you can have as many filter criteria as you like):



The Create Smart Bin dialog

- **Show in all projects checkbox:** Lets you create a persistent Smart Bin that appears in all projects in your project library. Smart Bins created this way will be found in the User Smart Bins folder inside every project's Smart Bin area in the Media Pool.
- **Match options:** For multi-criteria filtering, choosing All ensures that every single criteria you specify is met for a clip to be filtered. Choosing Any means that if only one out of several criteria is met, that clip will be filtered.

- **Filter criteria enable checkbox:** Lets you enable or disable any criteria without having to delete it.
- **Metadata category drop-down:** Lets you choose which category of metadata you want to select a criteria from. Each category of metadata that's available in the Metadata Editor is available from this drop-down menu. Additionally, Color Timeline Properties (containing many properties unique to the Color page timeline) and Media Pool Properties (containing every column in the Media Pool) provide access to additional metadata you can use for filtering.
- **Metadata type drop-down:** For choosing which exact type of metadata to use, of the options available in the selected metadata category.
- **Metadata criteria drop-down:** Lets you choose the criteria by which to filter, depending on the metadata you've selected. Options include "true/false," integer ranges, date ranges, string searches, flag and marker colors, et cetera.
- **Add filter criteria button:** Lets you add additional criteria to create multi-criteria filters. You could use multiple criteria to, for example, find all exterior clips, that also contain the keyword "Sunset," that aren't closeups, in order to find all the exterior long and medium shots in sunset lighting. Additionally, if you Option-click this button, you can add a nested match option in order to create even more sophisticated filters, such as when the filter must match all of one set of criteria, and any of another set of criteria.

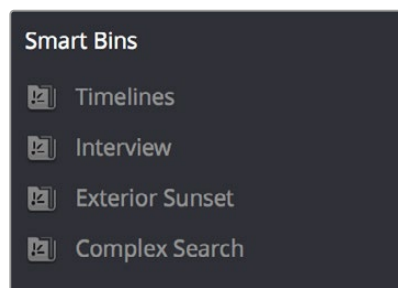


A complicated Smart Bin with multiple criteria and a second match option setting

As you're editing the filter criteria, the thumbnail timeline automatically updates to show you how the Smart Bin you're creating is working.

- 3 When you're done editing the filter criteria, click Create Smart Bin. The resulting Smart Bin appears in the Smart Bin area of the Bin list, at the left of the Media Pool's browser area.

Once you've created a Smart Bin, it appears in the lower half of the Media Pool's Bin list, alongside every other Smart Bin in that project. This keeps them organized, separate from the manually created bin shown above.



All Smart Bins appear together at the bottom of the Media Pool's Bin list

Once you've created a Smart Bin, you can re-edit it whenever the situation requires.

Methods of modifying existing Smart Bins:

- **To rename a Smart Bin:** Right-click the Smart Bin you want to rename, choose Rename from the contextual menu, enter a new name, and press Return.
- **To edit a Smart Bin:** Double-click the Smart Bin, then edit the filter criteria, and click OK.
- **To duplicate a Smart Bin:** Right-click any Smart Bin and choose Duplicate from the contextual menu. This is a good way to create multiple variations of a Smart Bin that you created with complex rules, where you need to create variations by modifying those rules without needing to reinvent the wheel each time.
- **To delete a Smart Bin:** Right-click the Smart Bin you want to delete, choose Delete Smart Bin from the contextual menu, and click Delete in the warning dialog. Deleting a Smart Bin does not delete any gathered media associated with that bin.

Smart Bins Work Better With Metadata

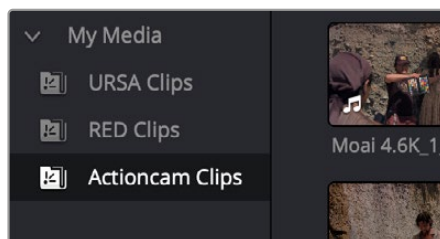
Keep in mind that the more metadata you associate with each clip, the more methods you have at your disposal for creating custom Smart Bins (for editing) and Smart Filters (for grading) with which to zero in on the clips you need for any given situation. This will not only make it easier to find what you need, but it'll help you to work faster. At the very least, it would be valuable for you to use the Metadata Editor to add information to each clip such as a Description, Shot and Scene designations, take information, and possibly some useful keywords such as character names, shot framing, interior or exterior keywords, and so on.

For example, if you've entered enough metadata, then you can create multi-criteria Smart Bins or Smart Filters that let you find the equivalent of "every close-up of Sally inside the diner," or "every long shot of Antonio outside in the parking lot." In a documentary, you could easily isolate "every interview shot of Louis from camera 1," or "every B-roll clip with Robyn." All of this will help you find media faster for editing, or quickly isolate similar clips that you need to match together for grading.

For more information about using the Metadata Editor, see *Chapter 19, "Using Clip Metadata."*

Organizing Smart Bins

Manually created Smart Bins can be organized into Folders and Sub-Folders for better sidebar management, just like regular bins.



Smart Bins organized into folders

To add a Smart Bin folder:

Right-click in the Smart Bins area and choose Add Folder from the contextual menu to create folders that you can drag Smart Bins into. Each folder has a disclosure triangle, so you can show or hide its contents.

Another benefit of Folders is that when you select a Folder, you can see the full contents of all Smart Bins inside of it in the Media Pool browsing area. Selecting any one Smart Bin then restricts the Media Pool to showing only the media reference by that Smart Bin.

Folders can be renamed, removed, opened as a new window, or sorted along with all other Smart Bins by right-clicking them and using commands in the contextual menu.

Duplicating Clips in the Media Pool

You can duplicate clips in order to create an instance of that media that's treated as a completely new source clip, entirely separate from the original instance of that clip that was imported into DaVinci Resolve. The duplicate is capable of storing individualized metadata and markers that are completely distinct from the original clip that was imported into your project.

To duplicate one or more clips:

- 1 Select one or more clips to duplicate.
- 2 Do one of the following:
 - Choose Edit > Duplicate Clip
 - Hold the Option key down while dragging one or more selected clips to another bin
 - Right-click a clip in the Media Pool, and choose Duplicate Clip from the Contextual Menu

Adding Clips From the Timeline to the Media Pool

You can also drag one or more clips from the Timeline back into the Media Pool to create a duplicate.

As with duplicating clips in the Media Pool, each duplicate is created as a new source clip that's entirely separate from the original instance of that clip that was imported into DaVinci Resolve and is capable of storing individualized metadata and markers that are completely distinct from the original clip that was imported into your project.

For example, the original clip in the Timeline remains conformed to the original clip that was first imported into the Media Pool; deleting the original clip from the Media Pool will make that clip “non-conformed” in the Timeline, while the duplicate you just created remains linked and available. If you're in this situation, you can always turn Conform Lock Enabled off for that clip in the Timeline and reconfirm the Timeline Clip to the duplicate you just created, but that's an extra step because the duplicate clip is considered by DaVinci Resolve to be a whole new piece of media that just happens to share the same clip details.

This may seem strange, but there are a variety of finishing workflows that use this capability, so it's good to know about.

Duplicating Timelines

Timelines can be duplicated for a variety of reasons: to create a backup of a timeline at a specific date, to create a variation of an edit, or to create separately graded versions.

To duplicate a Timeline:

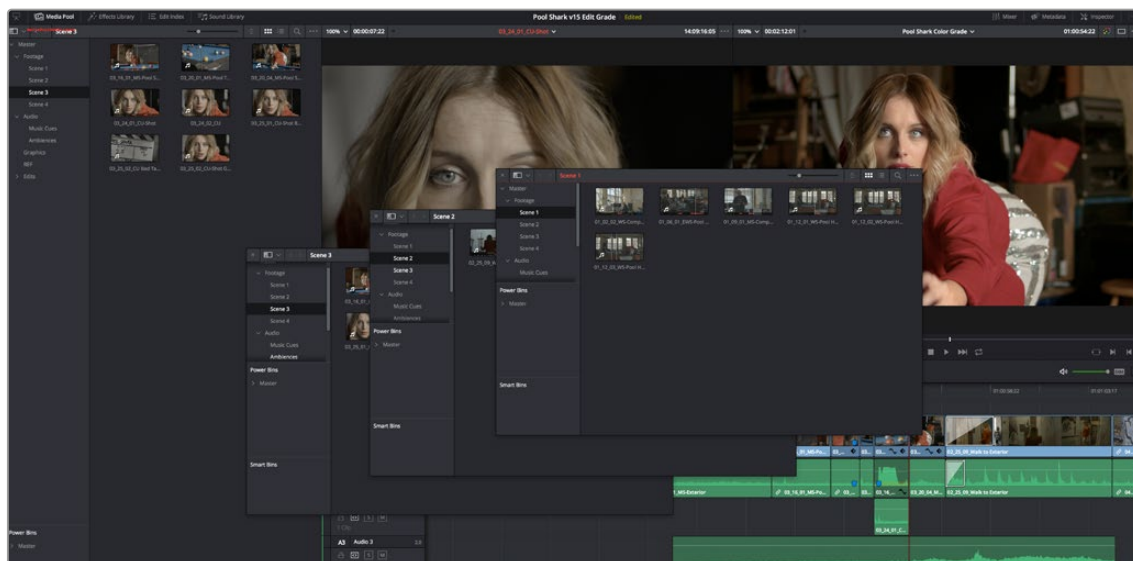
- Select a Timeline in the Media Pool, and choose Edit > Duplicate Timeline.
- Press Command-4 to give focus to the Timeline, and choose Edit > Duplicate Timeline.

Choosing How to Display Bins

Once you've created a bin structure for your project, you can customize how your bins are displayed, depending on how you like to work.

Showing Bins in Separate Windows

If you right-click a bin in the Bin list, you can choose “Open As New Window” to open that bin into its own window. That window is basically its own Media Pool, complete with its own Bin list, Power Bins and Smart Bins lists, and display controls.



Multiple Media Pool bins opened as new windows

When multiple Media Pool windows are open, the Workspace > Media Pool Windows submenu lets you bring a floating Media Pool window into focus when you have one or more open and hidden.

This is most useful when you have two displays connected to your workstation, as you can drag these separate bins to the second display while DaVinci Resolve is in single screen mode. If you hide the Bin list, not only do you get more room for clips, but you also prevent accidentally switching bins if you really want to only view a particular bin's contents in that window.

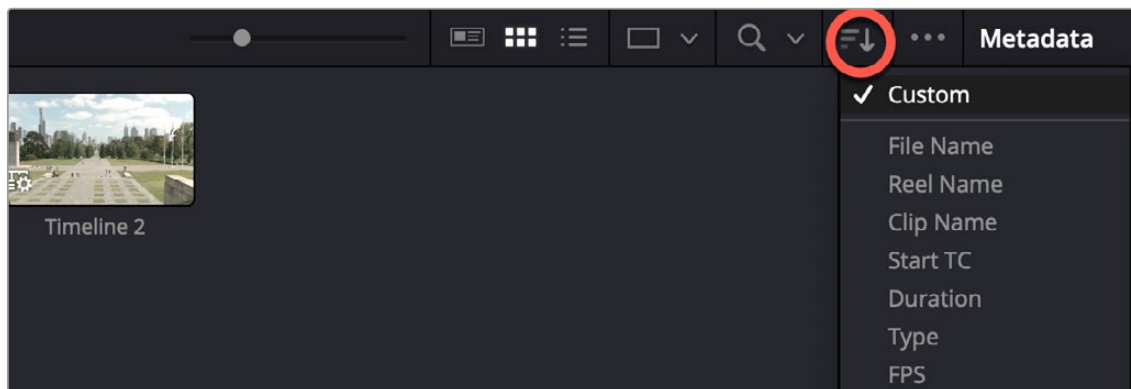
Using the Media Pool in Thumbnail View

If you work in Thumbnail view using the controls at the top right of the Media Pool, you have the option to resize the thumbnails to make them easier to see, and you can move the mouse pointer over each clip to hover scrub through its contents. Clicking any clip to select it displays it in the Media page Viewer. Whichever clip is currently selected is also output to video for monitoring.

In Thumbnail view, you can use the Sort Order drop-down, at the top right of the Media Pool, between the Icon Size slider and the Icon/List view buttons, to choose how clips are sorted. There are fourteen options: File Name, Reel Name, Clip Name, Start TC, Duration, Type, FPS, Audio Ch, Flags, Date Modified, Date Created, Shot, Scene, and Take.

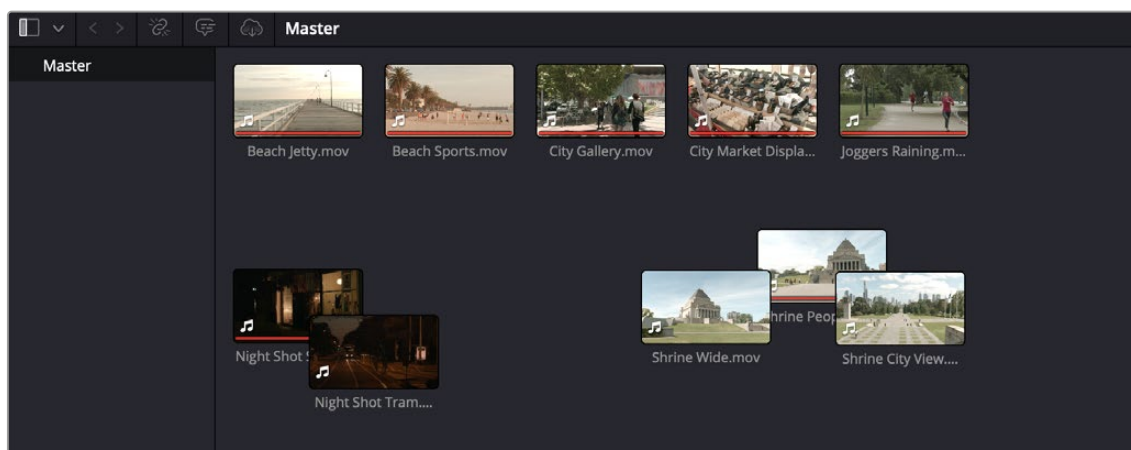
Custom Sort of Clip Thumbnails in the Media Pool

You can now manually reposition Media Pool clips in Thumbnail view, allowing you to group clips together in any way that visually makes sense to you.



You can turn on Custom Sort by selecting Custom in the Sort menu in the Media Pool.

Select Thumbnail View in the Media Pool, and from the sort menu select Custom. Then you can drag your thumbnails around in any order or grouping that you wish. If Snap to Grid is selected in the Sort menu, the icons will line up in neat rows and columns. Turn off Snap To Grid to enable freeform positioning. If two thumbnails are overlapping, the last one moved will show on top.



With Snap to Grid deselected you can rearrange thumbnails in any way (even overlapping), and it is retained throughout the Media Pool on all pages.

The position of your icon thumbnails are persistent in the Media Pools across all pages in DaVinci Resolve. The positioning is also retained if you temporarily choose another sort order, and then return to the Custom sort.

Working With Columns in List View

If you work in List view using the controls at the top right of the Media Pool, you gain additional organizational control by exposing columns that show the metadata that each clip contains, prior to media being added to your timeline. You can use these columns to help organize your media.

Methods of customizing metadata columns in List view:

- **To show or hide columns:** Right-click at the top of any column in the Media Pool to reveal the column list, and while the column list is open, click the checkboxes of any columns you want to show or hide. Unchecked columns cannot be seen. When you're finished, click anywhere else in the Media Pool to dismiss the column list.
- **To rearrange column order:** Drag any column header to the left or right to rearrange the column order.
- **To resize any column:** Drag the border between any two columns to the right or left to narrow or widen that column.
- **To sort by any column:** Click the column header you want to sort with. Each additional time you click, the same header toggles that column between ascending and descending sort order.

Once you've customized a column layout that works for your particular purpose, you can save it for future recall.

Methods of saving and using custom column layouts:

- **To create a column layout:** Show, hide, resize, and rearrange the columns you need for a particular task, then right-click any column header in the Media Pool, and choose Create Column Layout. Enter a name in the Create Column Layout dialog, and click OK.
- **To recall a column layout:** Right-click any column header in the Media Pool, choose the name of the column layout you want to use from the contextual menu, and choose Load from that item's submenu. All custom column layouts appear at the top of the list.
- **To edit a column layout:** Load the column layout you want to edit, make whatever changes you need to, then right-click any column header in the Media Pool, choose the name of the column layout you just edited from the contextual menu, and choose Update from that item's submenu.
- **To delete a column layout:** Right-click any column header in the Media Pool, choose the name of the column layout you want to delete from the contextual menu, and choose Delete from that item's submenu.

While the available columns of metadata correspond to those fields shown in the Metadata Editor, the available columns in the Media Pool of the Media and Edit pages are a subset of the total amount of metadata that's available, although they represent the most commonly used metadata you'll find yourself referring to when editing and finishing.

The available columns in List view include:

Angle: An editable field to contain the angle of the media in a multi-camera shoot.
Audio Bit Depth: The bit depth of any audio channels in the media file.
Audio Ch: The total number of audio tracks in the media file.
Audio Codec: The specific codec used by the audio portion of the media file.
Audio Offset: Lists the audio offset, in frames, for clips that have been synchronized to separately recorded audio. This parameter is editable in the Media Pool.
Bit Depth: The bit depth of the media file.
Camera #: The number assigned to a specific camera.
Clip Color: The current color assigned to that clip.

Clip Name: Editing the Clip Name lets you change the name with which clips appear throughout DaVinci Resolve when View > Use Clip Name for Clip Titles is enabled. By default, the clip name mirrors the source clip's file name. When editing the clip name in the List view of the Media Pool, you can use "metadata variables" that you can add as graphical tags that let you reference clip metadata. For example, you could add the corresponding metadata variable tags %scene_%shot_%take and that clip would display "12_A_3" as its name if "scene 12," "shot A," "take 3" were its metadata. The clip name can also be edited in the Clip Attributes window. For more information on the use of variables, as well as a list of all variables that are available in DaVinci Resolve, see <i>Chapter 16, "Using Variables and Keywords."</i>
Comments: A user-editable field for entering information about that clip.
Data Level: The data level setting for the media file.
Date Created: The date the media file was created.
Date Modified: The last date the media file was modified.
Description: A user-editable field for entering information about that clip.
Duration: The total duration of the clip, in timecode.
End: The last frame number of the media file.
End TC: The timecode value of the last frame in the media file.
FPS: The frame rate of the media file.
File Name: The name of the file on disk that clip is linked to.
File Path: The file path where that media file is located on disk.
Flags: Which flags, if any, have been added to a media file.
Format: The image format used by that clip, such as QuickTime, MXF, WAVE, and so on.
Frame/Field: Whether that media file is progressive or interlaced.
Frames: The total duration, in frames.
Good Take: An editable field to contain the circled state of media, relative to the script supervisor's notes.
H-FLIP: Whether that media file is horizontally flipped in DaVinci Resolve.
HDRX: Only displayed for R3D media, indicates whether or not it's HDRX media.
IDT: If ACES color science is selected in the Color Management panel of the Project Settings, the IDT used by that clip is listed here.
In: The timecode value of the In point, if any, that's stored for that clip.
Input Color Space: If Resolve Color Management is selected in the "Color Science" menu of the Color Management panel of the Project Settings, then this column will show the Input Color Space that has been assigned to each clip. By default, all clips inherit the Input Color Space setting that's been selected in the Color Management panel of the Project Settings.
Input LUT: Which input Lookup table has been assigned, if any.

Input Sizing Preset: The currently selected Input Format Preset, if there is one.
Keyword: A user-editable field for entering searchable keywords pertaining to that clip. Only shows clip keywords, not marker keywords.
Offline Reference: Lists the offline reference video that has been assigned to a given timeline.
Optimized Media: Populated with the resolution of whatever optimized media you've created (Original, Half, Quarter, and so on). Clips that have not been optimized appear with "None."
Out: The timecode value of the Out point, if any, that's stored for that clip.
PAR: The pixel aspect ratio, if assigned.
Reel Name: The reel name of that clip. Dynamically generated by the "Assist using reel names from the" setting in the General Options panel of the Project Settings.
Resolution: The frame size of the media file.
Roll/Card: An editable field to contain the roll number of media that was scanned from film.
S3D Sync: Shows a frame count when you slip an eye to fix non-synced timecode using the "Slip Opposite Eye One Frame Left/Right" commands. This parameter is editable in the Media Pool.
Sample Rate: The sample rate of the media file's audio, if there is any.
Scene: An editable field to contain the scene number of the media, relative to the script.
Shot: An editable field to contain the shot number of the media, relative to the scene.
Slate TC: The Slate timecode track used to sync audio with video.
Start: The first frame number of the media file.
Start KeyCode: The starting KeyCode value of a scanned negative.
Start TC: The timecode value of the first frame in the media file.
Take: An editable field to contain the take number of the media, relative to the shot.
Type: The type of item, such as Video+Audio, Video, Audio, Timeline, Multicam, Still, and so on.
Usage: After a timeline has been created by importing an AAF, EDL, or XML project, the Usage column automatically reflects how many times each clip is used in the project. This makes it easy to identify clips that aren't in use, and which can be removed from the Media Pool.
V-FLIP: Whether that media file is vertically flipped in DaVinci Resolve.
Video Codec: The specific codec used by the video portion of the media file.

Editable Name, Description, Keyword, and Comments Columns

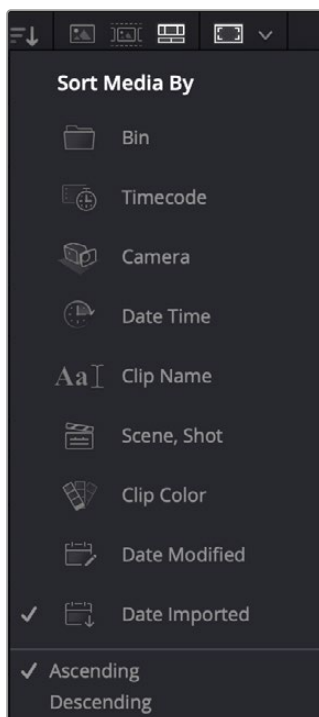
In addition to the Clip Name, when the Keyword, Description and Comments columns are displayed by the Media Pool in List view, you can edit their contents by clicking once within a clip's Description or Comments field, waiting a moment, and then clicking a second time to select that field.

Using Metadata View in the Media Pool

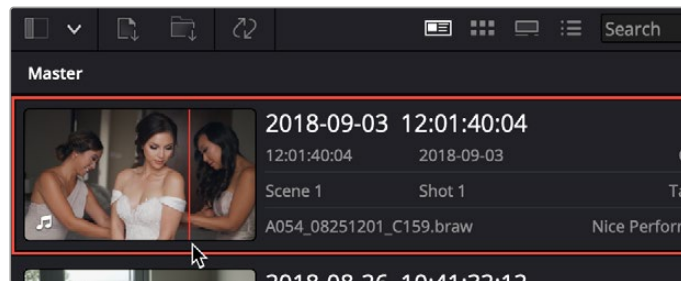
In the Metadata View mode, each clip is represented by its own card with a thumbnail and basic clip metadata information visible. This view is designed to have more metadata information than a thumbnail but more targeted information than the List view. This feature, combined with its sort modes, is a powerful way to organize and reorganize your clips in the Media Pool.

The metadata fields of the Metadata view (from the top down):

- **Thumbnail:** A scrubbable thumbnail image of your clip.
- **Row 1:** A main description field that is variable and determined by the sort order selection.
- **Row 2:** Start Timecode, Date Created, Camera #.
- **Row 3:** Scene, Shot, Take.
- **Row 4:** Clip Name, Comment.



The Media
Sort options



The Metadata View icon view (highlighted icon in the top bar),
showing the thumbnail being scrubbed next to the clip's metadata

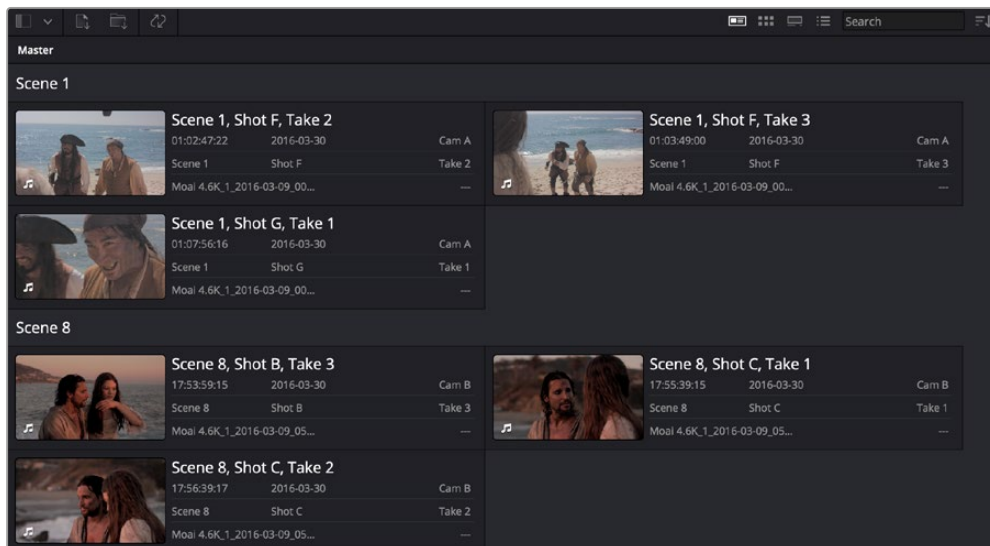
The strength of the Metadata view is the automatic clustering of your clips based on the sort order you choose in the Media Pool Sort By menu, at the very upper-right corner of the Media Pool.

Each different sort mode changes the main description field on the card, as well as re-arranging the Media Pool to reflect the selected organization method.

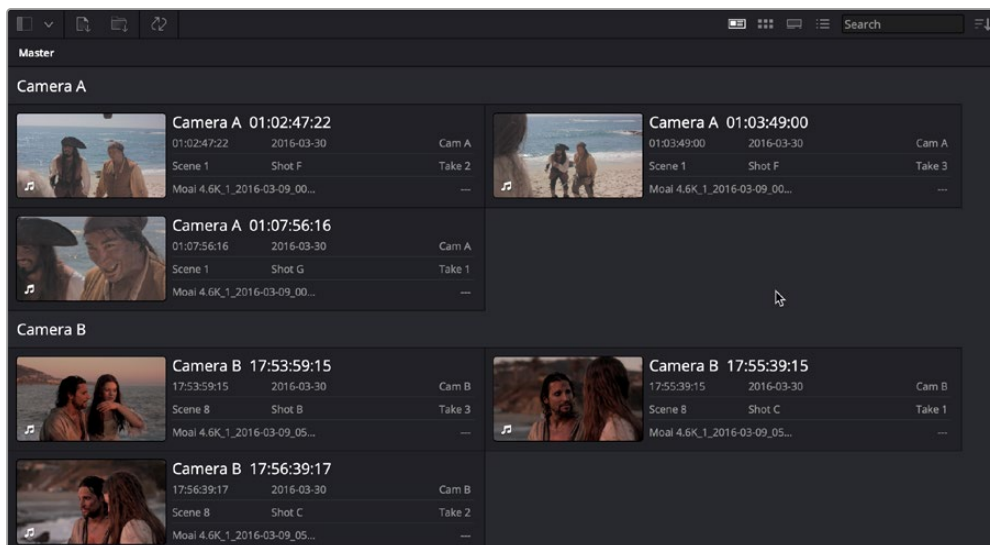
The sort modes available in the Metadata view are:

- **Bin:** This mode clusters the clips by bin, changes the main description field to clip name, and orders the list by timecode.
- **Timecode:** This mode clusters the clips by creation date, changes the main description field to creation date and start timecode, and orders the list by timecode.
- **Camera:** This mode clusters the clips by camera #, changes the main description field to camera # and start timecode, and orders the list by timecode.

- **Date Time:** This mode clusters the clips by day, changes the main description field to creation date and file name, and orders the list by timecode.
- **Clip Name:** This mode clusters the clips by the first letter of the clip name in alphabetical order, changes the main description field to clip name, and orders the list by timecode.
- **Scene, Shot:** This mode clusters the clips by scene, changes the main description field to scene-shot-take, and orders the list by scene-shot-take.
- **Clip Color:** This mode clusters the clips by clip color name, changes the main description field to creation date and start timecode, and orders the list by timecode.
- **Date Modified:** This mode clusters the clips by day, changes the main description field to creation date and file name, and orders the list by the last time the clip was modified by the OS filesystem.
- **Date Imported:** This mode clusters the clips by day, changes the main description field to creation date and file name, and orders the list by the date the clip was added to the Media Pool.
- **Ascending:** Orders the Media Pool from lowest numerical value to highest, and alphabetically from A to Z.
- **Descending:** Orders the Media Pool from highest numerical value to lowest, and alphabetically from Z to A.



The Metadata view with clips sorted by Scene-Shot-Take



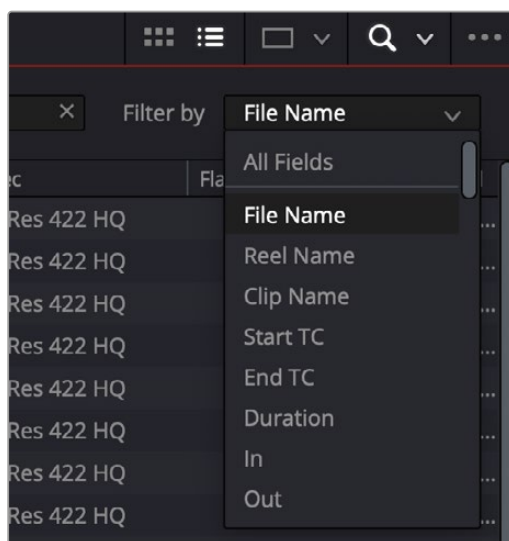
The Metadata view with the same clips sorted by Camera

Finding Clips, Timelines, and Media

There are several ways to locate different items in the Media Pool and Media Storage, be they clips, timelines, or media on disk.

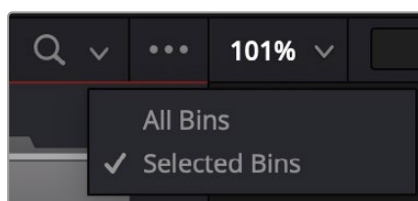
Finding Clips and/or Timelines Within the Media Pool

Clicking the magnifying glass button at the upper right-hand corner of the Media Pool exposes the Search Options, which by default can be used to locate one or more clips in the currently selected bin or bins, based on the metadata that's selected in the Filter By drop-down menu to the left of it.



The Search Options drop-down menu (as seen in the Edit page Media Pool) lets you choose what metadata you're searching

A drop-down menu right next to the magnifying glass icon lets you choose the scope of your search. This lets you choose whether a search looks through all bins in the current project for the specified criteria, or just looks at the currently open bin, or currently selected bins in the Bin list, in cases where you're looking for an instance of media in a specific hierarchical location of the Media Pool.



The drop-down menu next to the magnifying glass icon lets you set the bin search parameters

To find a clip in the Media Pool:

- 1 (Optional) Use the drop-down menu next to the Search button that exposes the Search and Filter by controls in the Media Pool to choose whether you select All Bins or Selected Bins.
- 2 (Optional) If you're searching Selected Bins, then open the Bin list and select one or more bins in which to search.

- 3 (Optional) Choose a criteria from the Search Options drop-down menu at the top right of the Media Pool; you can choose All Fields to do a simultaneous search of every metadata column in the Media Pool at once, or you can choose a specific criteria to restrict your search.
- 4 Type a search term in the Search field. As soon as you start typing, all clips that don't match the search criteria are temporarily hidden. To show all clips in the Media Pool again, click the cancel button at the right of the search field.

Finding Synced Audio

If you've synced dual-system audio and video clips together in DaVinci Resolve, you can find the audio clip that a video clip has been synced to using the following procedure.

To find the audio clip that a video clip has been synced to:

- Show the Media Pool in List view, and reference file name in the Synced Audio column.
- Right-click a video clip that's been synced to audio, and choose "Reveal synced audio in Media Pool" from the contextual menu. The bin holding the synced audio clip is opened and that clip is selected.

Finding Timeline Clips in the Media Pool

If you have a clip in a timeline and you want to find the corresponding clip that it's conformed to in the Media Pool, you can right-click that clip, and choose Find in Media Pool from the contextual menu.

Finding Timelines in the Media Pool

If you'd like to find the currently open timeline's location in the Media Pool, you can choose Timeline > Find Current Timeline in Media Pool.

Finding Media in the Media Storage Panel and Finder

If you find yourself needing to determine the location of a clip's source media file on disk, you can right-click an item in the Media Pool and choose Reveal in Media Storage panel. The Library automatically opens to the folder containing the media file you've selected, with that media file selected in the Library browser to the right.

Another feature that's only available for macOS systems is the ability to right-click an item in the Media Pool and choose Reveal In Finder. A file system window opens up, revealing the media file that clip is linked to.

Reveal Media Pool Bin from Multi-Bin or Search Displays

If you've searched for a clip and have results across multiple bins, you can now reveal where that clip is in the Media Pool by right-clicking on the clip and selecting Reveal Media Pool Bin from the contextual menu.

Going Immediately to a File System Location in the Media Browser

Conversely, if you drag a folder from the macOS Finder into the Media Storage panel, the Media Storage panel will immediately update to show the location of that folder.

Tracking Media Usage

As clips are added to timelines, two mechanisms come into play for keeping track of which clips are used in which timelines.

Thumbnail Clip Usage Indicators

Whenever you open a timeline, all thumbnails in the Media Pool automatically update to show highlighted usage bars to let you know which parts of that clip are used in that timeline.



Two colored highlights at the bottom of the thumbnail indicate which parts of a clip are used by the currently open timeline.

If you right-click on a thumbnail that shows usage, a Usage submenu shows you a list of each instance of that clip in the currently open timeline. Choosing an instance from this list jumps the playhead to that clip in the Timeline.

List View Clip Usage Column

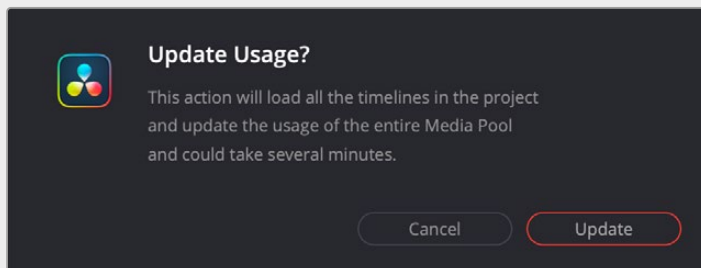
Exposing the Usage column when the Media Pool is in List view lets you see a value for the number of times a clip appears in the current timeline. This usage column is now automatically updated; no user intervention is required.

_2012-08-17_1409_C0004.mov		
Position	Usage	Frame
00:19:02	4	458
00:53:01	8	127
00:25:13	4	613
00:14:07		343
00:28:03		679

A Usage column shows how many times a clip is used in a timeline, after analysis.

Updating Clip Usage for all Timelines

You can also see which clips have been used across all the timelines in your project. From the Media Pool Options menu, select the “Update Usage for Entire Project” option.



The Update Usage dialog

DaVinci Resolve may prompt the user to load all timelines, if necessary, and update usage counts for Media Pool clips. This can take some time, depending on the complexity and number of timelines you have in your project.

Once completed, the usage count across all timelines will appear in the Usage column in the List view of the Media Pool.

NOTE: The Usage column increments for each clip item that appears in the Timeline. This means that if a clip consists of one video item and one video item linked together, the Usage column will show the number 2.

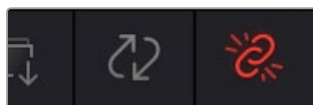
Relinking Media Simply

DaVinci Resolve keeps track of the relationship between clips in your project and their corresponding source media on disk. If, for whatever reason, source media that links to clips in your project becomes unavailable, DaVinci Resolve has several different methods of relinking those clips in the Media Pool.

This section summarizes the methods of relinking. For more comprehensive information on conforming projects and relinking media, see *Chapter 56, “Conforming and Relinking Clips.”*

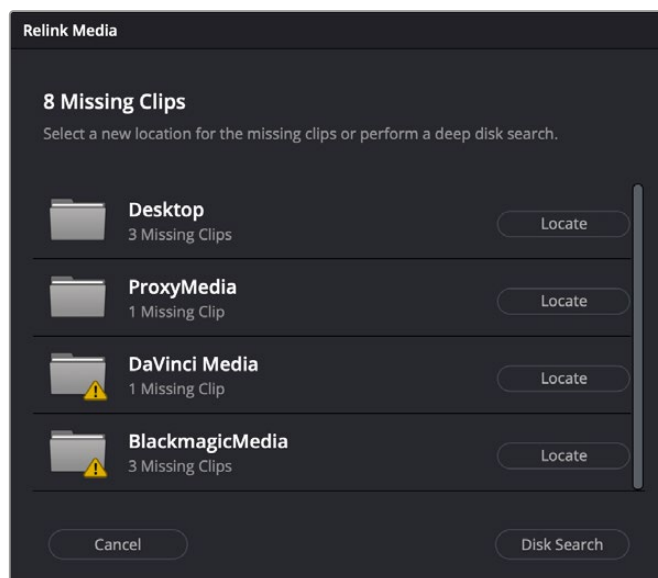
Relink Media

If DaVinci Resolve fails to find your media, a Relink Media icon in the Cut and Edit page’s Media Pool will highlight orange.



The Relink Media icon that appears for unlinked media

Clicking this icon opens a dialog box showing the volumes that the missing files initially belonged to. You can then use this information to track down the media on your file system, find that specific hard drive, or ask a client if they provided you the media from this volume. Clicking the Locate button lets you re-connect the missing clips to a new file location of your choosing. If the quick search initiated by the Locate buttons doesn't find media that you know is there, you can initialize an exhaustive deep disk search for the media by clicking on the Disk Search button.



The Relink Media dialog showing the volume names where the missing clips originated

Relink Selected Clips

The easiest method of relinking clips in your project that have gone offline is to use the appropriately named “Relink selected clips” command. This is the most flexible method of relinking clips in your project with clips in a file system directory of your choice, using file name and timecode as the primary criteria for drawing a correspondence between each clip and the corresponding media file on disk. When you relink clips this way, the original file path in DaVinci Resolve is ignored, so this is a good command to use to relink to media that’s been reorganized on disk.

To relink selected clips:

- 1 Do one of the following:
 - Select one or more clips in the Media Pool browser that you want to relink, then right-click one of the selected clips or the selected bin, and choose “Relink Selected Clips” from the contextual menu.
 - Select a bin in the Media Pool Bin list that contains clips you want to relink, then right-click one of the selected clips or the selected bin, and choose “Relink Clips for Selected Bin” from the contextual menu.
- 2 When the Relink File dialog opens, choose a directory in which to look for the files you want to relink to, and click OK. DaVinci Resolve attempts to find every clip with a matching file name in the subdirectories of the directory you chose, using the original file paths of the clips being relinked to do this as quickly as possible. By first looking for the clips in the directories they were originally in, relinking can be quite fast.

- 3 If there are any clips that couldn't be found using the method in step 2, you're prompted with the option to do a "deep search" by a second dialog. If you click Yes, then DaVinci Resolve will look for each clip inside every subdirectory of the directory you selected in step 2. This may take significantly longer, but it should be completely successful so long as the media that's required is within the selected directory structure.
- 4 If there are still other clips that couldn't be found, you're prompted to either choose another directory altogether to continue searching, or quit.

Change Source Folder

If you've used your file system to move media that's associated with a DaVinci Resolve project, but you haven't changed the directory structure with which it's organized, you can use the Change Source Folder command to quickly relink selected clips in the Media Pool to the new file path of the media on disk, using the original file paths as a guide. This is a good relinking method to use, if possible, for projects on a SAN where you don't want to risk the excessively long search times that could result from using the Relink command to examine a nested hierarchy of folders in a more flexible way.

To relink your Media Pool clips to a new location:

- 1 Select one or more clips in the Media Pool, then right-click one of the selected clips, and choose Change Source Folder from the contextual menu. The Relink Media window appears displaying the original path for the material, with controls for choosing a new directory.
- 2 Click the "Browse" button to the right of the Change To field, and then use the file navigation dialog to find the new location of the media file, select it, and click Open.
- 3 If you succeeded in finding the appropriate media file, click Change. Otherwise, click Cancel.